

# FMCG Demand Planning & Data Validation System

## From inconsistent data to confident decisions

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### Overview

This project demonstrates how to generate **reliable, actionable demand planning decisions from imperfect FMCG data**.

Instead of assuming clean inputs, the system:

- Validates data before use
  - Identifies inconsistencies
  - Introduces an operational correction loop (recounts)
  - Re-evaluates data reliability after intervention
  - Produces outputs that can be acted on with confidence
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### The Problem

In real business environments, planning data is rarely perfect.

This dataset included:

- Inconsistent product identifiers
- Multiple pack sizes distorting movement
- Formatting issues (scientific notation, decimals)
- No clear product hierarchy
- Movement data that did not reconcile cleanly

This creates a critical issue:

A correct calculation can still lead to a wrong decision.

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### Approach

Rather than forcing perfect data, I designed a system around **control and validation**:

1. Standardise movement into cases
  2. Rebuild product grouping logic
  3. Validate movement behaviour
  4. Classify reliability (Acceptable / Check / Problem)
  5. Isolate problem records
  6. Introduce recount process
  7. Apply corrected stock values
  8. Recalculate planning
  9. Revalidate outcomes
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### System Flow

Raw Data

→ Standardise  
→ Validate  
→ Flag Issues  
→ Recount  
→ Adjust Stock  
→ Recalculate  
→ Revalidate  
→ Decision

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### Dashboard Outputs

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#### 1. Data Health Overview

Shows:

- Total SKUs
- % Acceptable
- % Check
- % Problem

Current state:

- Acceptable: **63.64%**
- Check: **25.00%**
- Problem: **11.36%**

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## 2. Current Data Reliability

Visual distribution of data quality:

How much of the dataset can be trusted right now?

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## 3. Priority Actions

A focused action layer showing:

- Product (with pack size clarity)
- Suggested order
- Days of cover
- Data reliability status

This functions as a **working decision tool**, not just a report.

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## 4. Impact of Recount Process

Tracks improvement after intervention:

| Status     | Before | After  |
|------------|--------|--------|
| Acceptable | 27.27% | 63.64% |
| Check      | 42.05% | 25.00% |
| Problem    | 30.68% | 11.36% |

**Improvement: +36.36% reliable data**

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## Key Design Principles

- Do not plan on data you cannot trust
- Separate data issues from stock issues
- Preserve original data (no overwrites)
- Apply corrections as a layer
- Validate results after action
- Prioritise usability over complexity

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## Development Reality

This project was not a straight build.

Challenges included:

- Structured reference behaviour in Excel
- Named ranges vs tables conflicts
- Data type mismatches breaking lookups
- Formula debugging across dependent layers
- Balancing clean logic with usable outputs

Several parts were:

- built

- broken
- tested
- rebuilt

This reflects how real systems are developed.

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### Timeline

Built as an **iterative working session**:

| Phase | Activity                      |
|-------|-------------------------------|
| 1     | Data cleaning & structuring   |
| 2     | Product grouping logic        |
| 3     | Validation layer              |
| 4     | Debugging and restructuring   |
| 5     | Recount process design        |
| 6     | Feedback loop integration     |
| 7     | Dashboard build               |
| 8     | Final validation & refinement |

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### What This Demonstrates

- Demand Planning capability
  - Data validation under imperfect conditions
  - Operational thinking (not just analysis)
  - Ability to build a working correction process
  - Decision-making under uncertainty
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### Final Thought

This is not a perfect model.

It is a **realistic one**.

It shows how to move from:

Inconsistent data → Controlled decisions → Measurable improvement

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### Key Takeaway

Reliable decisions do not require perfect data — they require control, validation, and the ability to act on what can be trusted.